

Qiangfei Xia

professor

基本信息

姓名: Qiangfei Xia

所属院校: 马萨诸塞大学-阿默斯特

邮箱: qxia@umass.edu

地区: United States

学科领域

材料物理

人工智能

电子与计算机工程

微电子科学与工程

集成电路设计与集成系统

纳米材料与技术

电子科学与技术

研究领域

深度学习与神经网络

嵌入式系统与硬件设计

纳米物理与纳米器件

薄膜材料与表面物理

计算机体系结构

纳米电子与先进工艺

电路与系统设计

半导体器件物理与工艺

纳米电子学与光子学

存储器设计与架构

+5

个人简介

Dr. Qiangfei Xia is a distinguished professor in Electrical & Computer Engineering at the University of Massachusetts Amherst and the head of the Nanodevices and Integrated Systems Laboratory. He earned his Ph.D. in Electrical Engineering from Princeton University in 2007 and has been recognized with the Guggenheim Fellowship in Engineering. His research focuses on beyond-CMOS devices, integrated systems, and enabling technologies, with applications in machine intelligence, reconfigurable RF systems, and hardware security. Notably, Dr. Xia has developed a reliable multilevel resistance-switching device and the world's smallest memristive

devices, contributing significantly to the field of in-memory computing and artificial neural networks.

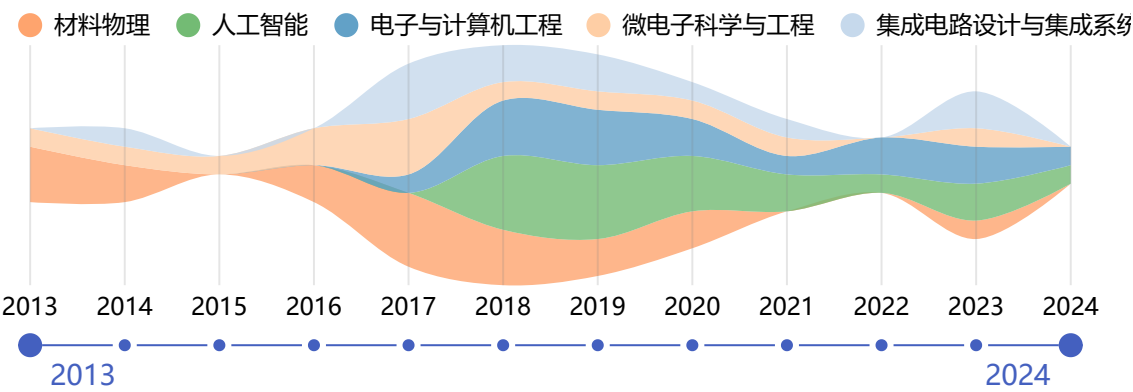
工作经历

UMass Amherst - Assistant Professor	2010-10 - 2016-01
UMass Amherst - Associate Professor	2016-01 - 2018-09
UMass Amherst - Professor	2018-09 - 至今
Hewlett Packard Laboratories - Research Associate	2007 - 2010-10

教育经历

Princeton University - PhD	null - 2007
Shanghai Jiao Tong University - MS	null - 2001
Shanghai Jiao Tong University - BE	null - 1998

近年发表的论文



MERRC: A Memristor-Enabled Reconfigurable Low-Power Reservoir Computing Architecture at the Edge
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS
Nowshin, Fabiha; Huang, Yi; Sarkar, Md. Rubel; Xia, Qiangfei; Yi, Yang
被引用次数: 4 2024

Diffusive Memristors with Uniform and Tunable Relaxation Time for Spike Generation in Event-Based Pattern Recognition
ADVANCED MATERIALS
Ye, Fan; Kiani, Fatemeh; Huang, Yi; Xia, Qiangfei

被引用次数: 64

2023

Memristive Field-Programmable Analog Arrays for Analog Computing

ADVANCED MATERIALS

Li, Yunning; Song, Wenhao; Wang, Zhongrui; Jiang, Hao; Yan, Peng; Lin, Peng; Li, Can; Rao, Mingyi; Barnell, Mark; Wu, Qing; Ganguli, Sabyasachi; Roy, Ajit K.; Xia, Qiangfei; Yang, J. Joshua

被引用次数: 10

2023

Memristor devices denoised to achieve thousands of conductance levels

NATURE

Yang, J. Joshua; Xia, Qiangfei

被引用次数: 0

2023

Thousands of conductance levels in memristors integrated on CMOS

NATURE

Rao, Mingyi; Tang, Hao; Wu, Jiangbin; Song, Wenhao; Zhang, Max; Yin, Wenbo; Zhuo, Ye; Kiani, Fatemeh; Chen, Benjamin; Jiang, Xiangqi; Liu, Hefei; Chen, Hung-Yu; Midya, Rivu; Ye, Fan; Jiang, Hao; Wang, Zhongrui; Wu, Mingche; Hu, Miao; Wang, Han; Xia, Qiangfei; Ge, Ning; Li, Ju; Yang, J. Joshua

被引用次数: 147

2023

Bipolar flash for bifunctional computing operations

NATURE NANOTECHNOLOGY

Kiani, Fatemeh; Xia, Qiangfei

被引用次数: 0

2023

From memristive devices to neuromorphic systems

APPLIED PHYSICS LETTERS

Huang, Yi; Kiani, Fatemeh; Ye, Fan; Xia, Qiangfei

被引用次数: 7

2023

A Dynamical Compact Model of Diffusive and Drift Memristors for Neuromorphic Computing

ADVANCED ELECTRONIC MATERIALS

Zhuo, Ye; Midya, Rivu; Song, Wenhao; Wang, Zhongrui; Asapu, Shiva; Rao, Mingyi; Lin, Peng; Jiang, Hao; Xia, Qiangfei; Williams, R. Stanley; Yang, J. Joshua

被引用次数: 25

2022

The secret order of disorder

NATURE MATERIALS

Xia, Qiangfei; Yang, J. Joshua; Midya, Rivu

被引用次数: 3

2022

A fully hardware-based memristive multilayer neural network

SCIENCE ADVANCES

Kiani, Fatemeh; Yin, Jun; Wang, Zhongrui; Yang, J. Joshua; Xia, Qiangfei

被引用次数: 45

2021

Standards for the Characterization of Endurance in Resistive Switching Devices

ACS NANO

Lanza, Mario; Waser, Rainer; Ielmini, Daniele; Yang, J. Joshua; Goux, Ludovic; Sune, Jordi; Kenyon, Anthony Joseph; Mehonic, Adnan; Spiga, Sabina; Rana, Vikas; Wiefels, Stefan; Menzel, Stephan; Valov, Ilia; Villena, Marco A.; Miranda, Enrique; Jing, Xu; Campabadal, Francesca; Gonzalez, Mireia B.; Aguirre, Fernando; Palumbo, Felix; Zhu, Kaichen; Roldan, Juan Bautista; Puglisi, Francesco Maria; Larcher, Luca; Hou, Tuo-Hung; Prodromakis, Themis; Yang, Yuchao; Huang, Peng; Wan, Tianqing; Chai, Yang; Pey, Kin Leong; Raghavan, Nagarajan; Duenas, Salvador; Wang, Tao; Xia, Qiangfei; Pazos, Sebastian

被引用次数: 137

2021

Roadmap on emerging hardware and technology for machine learning

NANOTECHNOLOGY

Berggren, Karl; Xia, Qiangfei; Likharev, Konstantin K.; Strukov, Dmitri B.; Jiang, Hao; Mikolajick, Thomas; Querlioz, Damien; Salinga, Martin; Erickson, John R.; Pi, Shuang; Xiong, Feng; Lin, Peng; Li, Can; Chen, Yu; Xiong, Shisheng; Hoskins, Brian D.; Daniels, Matthew W.; Madhavan, Advait; Liddle, James A.; McClelland, Jabez J.; Yang, Yuchao; Rupp, Jennifer; Nonnenmann, Stephen S.; Cheng, Kwang-Ting; Gong, Nanbo; Lastras-Montano, Miguel Angel; Talin, A. Alec; Salleo, Alberto; Shastri, Bhavin J.; de Lima, Thomas Ferreira; Prucnal, Paul; Tait, Alexander N.; Shen, Yichen; Meng, Huaiyu; Roques-Carmes, Charles; Cheng, Zengguang; Bhaskaran, Harish; Jariwala, Deep; Wang, Han; Shainline, Jeffrey M.; Segall, Kenneth; Yang, J. Joshua; Roy, Kaushik; Datta, Suman; Raychowdhury, Arijit

被引用次数: 112

2021

Power-efficient combinatorial optimization using intrinsic noise in memristor Hopfield neural networks

NATURE ELECTRONICS

Cai, Fuxi; Kumar, Suhas; Van Vaerenbergh, Thomas; Sheng, Xia; Liu, Rui; Li, Can; Liu, Zhan; Foltin, Martin; Yu, Shimeng; Xia, Qiangfei; Yang, J. Joshua; Beausoleil, Raymond; Lu, Wei D.; Strachan, John Paul

被引用次数: 236

2020

A Low-Current and Analog Memristor with Ru as Mobile Species

ADVANCED MATERIALS

Yoon, Jung Ho; Zhang, Jiaming; Lin, Peng; Upadhyay, Navnidhi; Yan, Peng; Liu, Yuzi; Xia, Qiangfei; Yang, J. Joshua

被引用次数: 73

2020

A Memristor with Low Switching Current and Voltage for 1S1R Integration and Array Operation

ADVANCED ELECTRONIC MATERIALS

Upadhyay, Navnidhi K.; Sun, Wen; Lin, Peng; Joshi, Saumil; Midya, Rivu; Zhang, Xumeng; Wang, Zhongrui; Jiang, Hao; Yoon, Jung Ho; Rao, Mingyi; Chi, Miaofang; Xia, Qiangfei; Yang, J. Joshua

被引用次数: 64

2020

Three-dimensional memristor circuits as complex neural networks

NATURE ELECTRONICS

Lin, Peng; Li, Can; Wang, Zhongrui; Li, Yunning; Jiang, Hao; Song, Wenhao; Rao, Mingyi; Zhuo, Ye; Upadhyay, Navnidhi K.; Barnell, Mark; Wu, Qing; Yang, J. Joshua; Xia, Qiangfei

被引用次数: 258

2020

Resistive switching materials for information processing

NATURE REVIEWS MATERIALS

Wang, Zhongrui; Wu, Huaqiang; Burr, Geoffrey W.; Hwang, Cheol Seong; Wang, Kang L.; Xia, Qiangfei; Yang, J. Joshua

被引用次数: 704

2020

Committee machines-a universal method to deal with non-idealities in memristor-based neural networks

NATURE COMMUNICATIONS

Joksas, D.; Freitas, P.; Chai, Z.; Ng, W. H.; Buckwell, M.; Li, C.; Zhang, W. D.; Xia, Q.; Kenyon, A. J.; Mehonic, A.

被引用次数: 52

2020

An artificial spiking afferent nerve based on Mott memristors for neurorobotics

NATURE COMMUNICATIONS

Zhang, Xumeng; Zhuo, Ye; Luo, Qing; Wu, Zuheng; Midya, Rivu; Wang, Zhongrui; Song, Wenhao; Wang, Rui; Upadhyay, Navnidhi K.; Fang, Yilin; Kiani, Fatemeh; Rao, Mingyi; Yang, Yang; Xia, Qiangfei; Liu, Qi; Liu, Ming; Yang, J. Joshua

被引用次数: 254

2020

Emerging Memory Devices for Neuromorphic Computing

ADVANCED MATERIALS TECHNOLOGIES

Upadhyay, Navnidhi K.; Jiang, Hao; Wang, Zhongrui; Asapu, Shiva; Xia, Qiangfei; Yang, J. Joshua

被引用次数: 348

2019

Artificial Neural Network (ANN) to Spiking Neural Network (SNN) Converters Based on Diffusive Memristors

ADVANCED ELECTRONIC MATERIALS

Midya, Rivu; Wang, Zhongrui; Asapu, Shiva; Joshi, Saumil; Li, Yunning; Zhuo, Ye; Song, Wenhao; Jiang, Hao; Upadhyay, Navnidhi; Rao, Mingyi; Lin, Peng; Li, Can; Xia, Qiangfei; Yang, J. Joshua

被引用次数: 100

2019

Scalable 3D Ta:SiO_x Memristive Devices

ADVANCED ELECTRONIC MATERIALS

Jiang, Hao; Li, Can; Lin, Peng; Pi, Shuang; Yang, Jianhua; Joshua, Xia, Qiangfei

被引用次数: 5

2019

Reinforcement learning with analogue memristor arrays

NATURE ELECTRONICS

Wang, Zhongrui; Li, Can; Song, Wenhao; Rao, Mingyi; Belkin, Daniel; Li, Yunning; Yan, Peng; Jiang, Hao; Lin, Peng; Hu, Miao; Strachan, John Paul; Ge, Ning; Barnell, Mark; Wu, Qing; Bartos, Andrew G.; Qiu, Qinru; Williams, R. Stanley; Xia, Qiangfei; Yang, J. Joshua

被引用次数: 269

2019

Artificial Neural Network Based on Doped HfO₂ Ferroelectric Capacitors With Multilevel Characteristics

IEEE ELECTRON DEVICE LETTERS

Zheng, Qilin; Wang, Zongwei; Gong, Nanbo; Yu, Zhizhen; Chen, Cheng; Cai, Yimao; Huang, Qianqian; Jiang, Hao; Xia, Qiangfei; Huang, Ru

被引用次数: 49

2019

Memristive crossbar arrays for brain-inspired computing

NATURE MATERIALS

Xia, Qiangfei; Yang, J. Joshua

被引用次数: 1227

2019

Memristor crossbar arrays with 6-nm half-pitch and 2-nm critical dimension

NATURE NANOTECHNOLOGY

Pi, Shuang; Li, Can; Jiang, Hao; Xia, Weiwei; Xin, Huolin; Yang, J. Joshua; Xia, Qiangfei

被引用次数: 407

2019

Understanding memristive switching via in situ characterization and device modeling

NATURE COMMUNICATIONS

Sun, Wen; Gao, Bin; Chi, Miaofang; Xia, Qiangfei; Yang, J. Joshua; Qian, He; Wu, Huaqiang

被引用次数: 273

2019

Memristor-Based Analog Computation and Neural Network Classification with a Dot Product Engine

ADVANCED MATERIALS

Hu, Miao; Graves, Catherine E.; Li, Can; Li, Yunning; Ge, Ning; Montgomery, Eric; Davila, Noraica; Jiang, Hao; Williams, R. Stanley; Yang, J. Joshua; Xia, Qiangfei; Strachan, John Paul

被引用次数: 601

2018

Artificial neural networks based on memristive devices

SCIENCE CHINA-INFORMATION SCIENCES

Ravichandran, Vignesh; Li, Can; Banagozar, Ali; Yang, J. Joshua; Xia, Qiangfei

被引用次数: 26

2018

Efficient and self-adaptive in-situ learning in multilayer memristor neural networks

NATURE COMMUNICATIONS

Li, Can; Belkin, Daniel; Li, Yunning; Yan, Peng; Hu, Miao; Ge, Ning; Jiang, Hao; Montgomery, Eric; Lin, Peng; Wang, Zhongrui; Song, Wenhao; Strachan, John Paul; Barnell, Mark; Wu, Qing; Williams, R. Stanley; Yang, J. Joshua; Xia, Qiangfei

被引用次数: 453

2018

Silicon Oxide (SiO_x): A Promising Material for Resistance Switching?

ADVANCED MATERIALS

Mehonic, Adnan; Shluger, Alexander L.; Gao, David; Valov, Ilia; Miranda, Enrique; Ielmini, Daniele; Bricalli, Alessandro; Ambrosi, Elia; Li, Can; Yang, J. Joshua; Xia, Qiangfei; Kenyon, Anthony J.

被引用次数: 160

2018

Nucleation limited switching (NLS) model for HfO₂-based metal-ferroelectric-metal (MFM) capacitors: Switching kinetics and retention characteristics

APPLIED PHYSICS LETTERS

Gong, N.; Sun, X.; Jiang, H.; Chang-Liao, K. S.; Xia, Q.; Ma, T. P.

被引用次数: 90

2018

Tutorial: Fabrication and three-dimensional integration of nanoscale memristive devices and

JOURNAL OF APPLIED PHYSICS

Lin, Peng; Xia, Qiangfei

被引用次数: 12

2018

Nanoscale diffusive memristor crossbars as physical unclonable functions

NANOSCALE

Zhang, R.; Jiang, H.; Wang, Z. R.; Lin, P.; Zhuo, Y.; Holcomb, D.; Zhang, D. H.; Yang, J. J.; Xia, Q.

被引用次数: 56

2018

An artificial nociceptor based on a diffusive memristor

NATURE COMMUNICATIONS

Yoon, Jung Ho; Wang, Zhongrui; Kim, Kyung Min; Wu, Huaqiang; Ravichandran, Vignesh; Xia, Qiangfei; Hwang, Cheol Seong; Yang, J. Joshua

被引用次数: 324

2018

Review of memristor devices in neuromorphic computing: materials sciences and device challenges

JOURNAL OF PHYSICS D-APPLIED PHYSICS

Li, Yibo; Wang, Zhongrui; Midya, Rivu; Xia, Qiangfei; Yang, J. Joshua

被引用次数: 381

2018

Threshold Switching of Ag or Cu in Dielectrics: Materials, Mechanism, and Applications

ADVANCED FUNCTIONAL MATERIALS

Wang, Zhongrui; Rao, Mingyi; Midya, Rivu; Joshi, Saumil; Jiang, Hao; Lin, Peng; Song, Wenhao; Asapu, Shiva; Zhuo, Ye; Li, Can; Wu, Huaqiang; Xia, Qiangfei; Yang, J. Joshua

被引用次数: 264

2018

Capacitive neural network with neuro-transistors

NATURE COMMUNICATIONS

Wang, Zhongrui; Rao, Mingyi; Han, Jin-Woo; Zhang, Jiaming; Lin, Peng; Li, Yunning; Li, Can; Song, Wenhao; Asapu, Shiva; Midya, Rivu; Zhuo, Ye; Jiang, Hao; Yoon, Jung Ho; Upadhyay, Navnidhi Kumar; Joshi, Saumil; Hu, Miao; Strachan, John Paul; Barnell, Mark; Wu, Qing; Wu, Huaqiang; Qiu, Qinru; Williams, R. Stanley; Xia, Qiangfei; Yang, J. Joshua

被引用次数: 186

2018

Ferroelectric transistors with monolayer molybdenum disulfide and ultra-thin aluminum-doped hafnium oxide

APPLIED PHYSICS LETTERS

Yap, Wui Chung; Jiang, Hao; Liu, Jialun; Xia, Qiangfei; Zhu, Wenjuan

被引用次数: 46

2017

Three-dimensional crossbar arrays of self-rectifying Si/SiO₂/Si memristors

NATURE COMMUNICATIONS

Li, Can; Han, Lili; Jiang, Hao; Jang, Moon-Hyung; Lin, Peng; Wu, Qing; Barnell, Mark; Yang, J. Joshua; Xin, Huolin L.; Xia, Qiangfei

被引用次数: 133

2017

Truly Electroforming-Free and Low-Energy Memristors with Preconditioned Conductive Tunneling Paths

ADVANCED FUNCTIONAL MATERIALS

Yoon, Jung Ho; Zhang, Jiaming; Ren, Xiaochen; Wang, Zhongrui; Wu, Huaqiang; Li, Zhiyong; Barnell, Mark; Wu, Qing; Lauhon, Lincoln J.; Xia, Qiangfei; Yang, J. Joshua

被引用次数: 79

2017

A novel true random number generator based on a stochastic diffusive memristor

NATURE COMMUNICATIONS

Jiang, Hao; Belkin, Daniel; Savel'ev, Sergey E.; Lin, Siyan; Wang, Zhongrui; Li, Yunning; Joshi, Saumil; Midya, Rivu; Li, Can; Rao, Mingyi; Barnell, Mark; Wu, Qing; Yang, J. Joshua; Xia, Qiangfei

被引用次数: 187

2017

ORGANIC ELECTRONICS Battery-like artificial synapses

NATURE MATERIALS

Yang, J. Joshua; Xia, Qiangfei

被引用次数: 33

2017

Efficient electrical control of thin-film black phosphorus bandgap

NATURE COMMUNICATIONS

Deng, Bingchen; Tran, Vy; Xie, Yujun; Jiang, Hao; Li, Cheng; Guo, Qiushi; Wang, Xiaomu; Tian, He; Koester, Steven J.; Wang, Han; Cha, Judy J.; Xia, Qiangfei; Yang, Li; Xia, Fengnian

被引用次数: 284

2017

An efficient analog Hamming distance comparator realized with a unipolar memristor array: a showcase of physical computing

SCIENTIFIC REPORTS

Ge, Ning; Yoon, Jung Ho; Hu, Miao; Merced-Grafals, E. J.; Davila, Noraica; Strachan, John Paul; Li, Zhiyong; Holder, Helen; Xia, Qiangfei; Williams, R. Stanley; Zhou, Xing; Yang, J. Joshua

被引用次数: 27

2017

Anatomy of Ag/Hafnia-Based Selectors with 10¹⁰ Nonlinearity

ADVANCED MATERIALS

Midya, Rivu; Wang, Zhongrui; Zhang, Jiaming; Savel'ev, Sergey E.; Li, Can; Rao, Mingyi; Jang, Moon Hyung; Joshi, Saumil; Jiang, Hao; Lin, Peng; Norris, Kate; Ge, Ning; Wu, Qing; Barnell, Mark; Li, Zhiyong; Xin, Huolin L.; Williams, R. Stanley; Xia, Qiangfei; Yang, J. Joshua

被引用次数: 309

2017

Memristors with diffusive dynamics as synaptic emulators for neuromorphic computing

NATURE MATERIALS

Wang, Zhongrui; Joshi, Saumil; Savel'ev, Sergey E.; Jiang, Hao; Midya, Rivu; Lin, Peng; Hu, Miao; Ge, Ning; Strachan, John Paul; Li, Zhiyong; Wu, Qing; Barne, Mark; Li, Geng-Lin; Xin, Huolin L.; Williams, R. Stanley; Xia, Qiangfei; Yang, J. Joshua

被引用次数: 1813

2017

Sub-10 nm Ta Channel Responsible for Superior Performance of a HfO₂ Memristor

SCIENTIFIC REPORTS

Jiang, Hao; Han, Lili; Lin, Peng; Wang, Zhongrui; Jang, Moon Hyung; Wu, Qing; Barnell, Mark; Yang, J. Joshua; Xin, Huolin L.; Xia, Qiangfei

被引用次数: 176

2016

A Dynamically Reconfigurable Ambipolar Black Phosphorus Memory Device

ACS NANO

Tian, He; Deng, Bingchen; Chin, Matthew L.; Yan, Xiaodong; Jiang, Hao; Han, Shu-Jen; Sun, Vivian; Xia, Qiangfei; Dubey, Madan; Xia, Fengnian; Wang, Han

被引用次数: 99

2016

Fabrication of sub-10 nm metal nanowire arrays with sub-1 nm critical dimension control

NANOTECHNOLOGY

Pi, Shuang; Lin, Peng; Xia, Qiangfei

被引用次数: 14

2016

Electrochemical metallization switching with a platinum group metal in different oxides

NANOSCALE

Wang, Zhongrui; Jiang, Hao; Jang, Moon Hyung; Lin, Peng; Ribbe, Alexander; Xia, Qiangfei; Yang, J. Joshua

被引用次数: 37

2016

Electrochemical metallization switching with a platinum group metal in different oxides (vol 8, pg 11766, 2016)

NANOSCALE

Wang, Zhongrui; Jiang, Hao; Jang, Moon Hyung; Lin, Peng; Ribbe, Alexander; Wu, Qing; Barnell, Mark; Xia, Qiangfei; Yang, J. Joshua

被引用次数: 2

2016

Synthetic Biological Protein Nanowires with High Conductivity

SMALL

Tan, Yang; Adhikari, Ramesh Y.; Malvankar, Nikhil S.; Pi, Shuang; Ward, Joy E.; Woodard, Trevor L.; Nevin, Kelly P.; Xia, Qiangfei; Tuominen, Mark T.; Lovley, Derek R.

被引用次数: 139

2016

Black Phosphorus Mid-Infrared Photodetectors with High Gain

NANO LETTERS

Guo, Qiushi; Pospischil, Andreas; Bhuiyan, Maruf; Jiang, Hao; Tian, He; Farmer, Damon; Deng, Bingchen; Li, Cheng; Han, Shu-Jen; Wang, Han; Xia, Qiangfei; Ma, Tso-Ping; Mueller, Thomas; Xia, Fengnian

被引用次数: 684

2016

Nanoimprint lithography 20 years on

NANOTECHNOLOGY

Xia, Qiangfei; Pease, R. Fabian

被引用次数: 11	2015
Nanoscale memristive radiofrequency switches	
NATURE COMMUNICATIONS	
Pi, Shuang; Ghadiri-Sadrabadi, Mohammad; Bardin, Joseph C.; Xia, Qiangfei	
被引用次数: 110	2015
3D integration of planar crossbar memristive devices with CMOS substrate	
NANOTECHNOLOGY	
Lin, Peng; Pi, Shuang; Xia, Qiangfei	
被引用次数: 28	2014
Concentric rings of polystyrene and titanium dioxide nanoparticles patterned by alternating current signal guided coffee ring effect	
APPLIED PHYSICS LETTERS	
Mu, Jinhua; Lin, Peng; Xia, Qiangfei	
被引用次数: 10	2014
Single- and bi-layer memristive devices with tunable properties using TiOx switching layers deposited by reactive sputtering	
APPLIED PHYSICS LETTERS	
Jiang, Hao; Xia, Qiangfei	
被引用次数: 12	2014
Black Phosphorus Radio-Frequency Transistors	
NANO LETTERS	
Wang, Han; Wang, Xiaomu; Xia, Fengnian; Wang, Luhao; Jiang, Hao; Xia, Qiangfei; Chin, Matthew L.; Dubey, Madan; Han, Shu-jen	
被引用次数: 330	2014
Mold cleaning with polydimethylsiloxane for nanoimprint lithography	
NANOTECHNOLOGY	
Lin, Peng; Pi, Shuang; Jiang, Hao; Xia, Qiangfei	
被引用次数: 6	2013
Improvement of resistive switching uniformity for TiO2-based memristive devices by introducing a thin HfO2 layer	
JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B	
Jiang, Hao; Xia, Qiangfei	
被引用次数: 11	2013
Cross point arrays of 8 nm x 8 nm memristive devices fabricated with nanoim print lithography	
JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B	
Pi, Shuang; Lin, Peng; Xia, Qiangfei	
被引用次数: 82	2013

Effect of voltage polarity and amplitude on electroforming of TiO2 based memristive devices

NANOSCALE

Jiang, Hao; Xia, Qiangfei

被引用次数: 17

2013

Low voltage resistive switching devices based on chemically produced silicon oxide

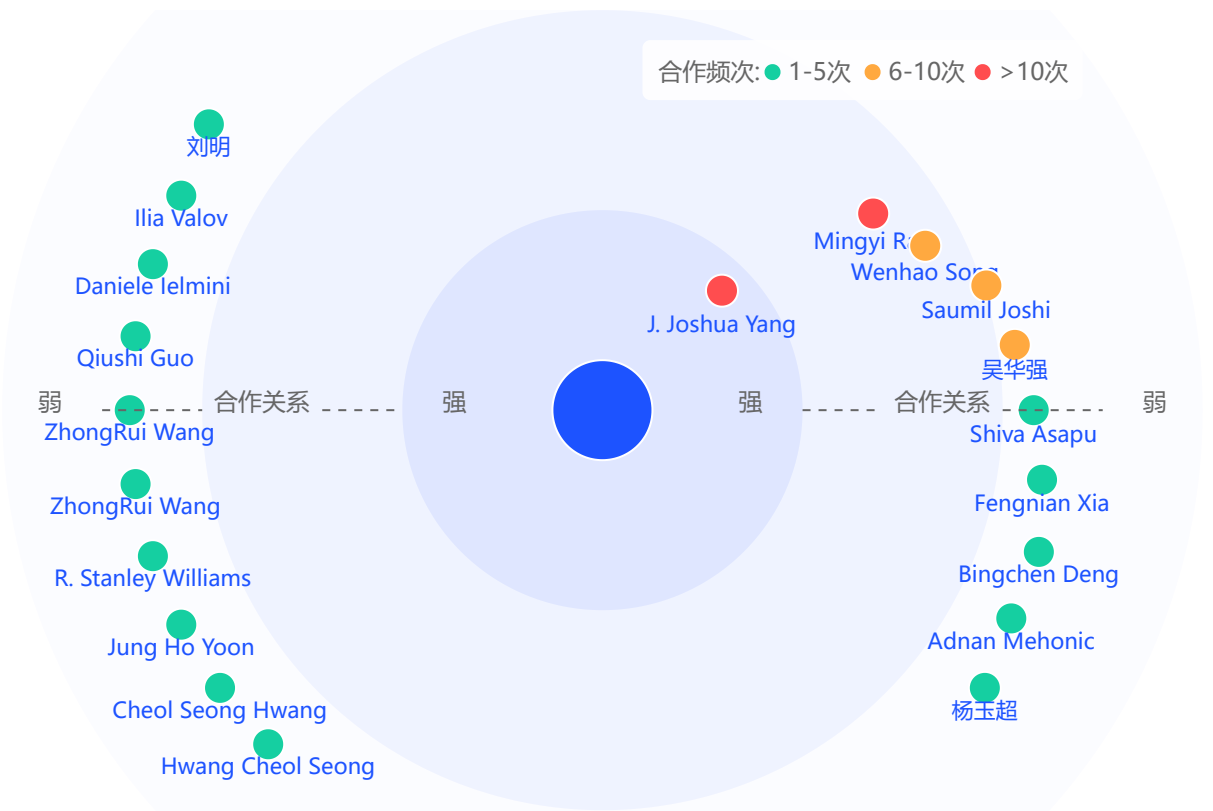
APPLIED PHYSICS LETTERS

Li, Can; Jiang, Hao; Xia, Qiangfei

被引用次数: 31

2013

学术合作



J. Joshua Yang	合作次数: 24
马萨诸塞大学-阿默斯特	
Mingyi Rao	合作次数: 12
马萨诸塞大学-阿默斯特	
Wenhao Song	合作次数: 10
南加州大学	
Saumil Joshi	合作次数: 7
马萨诸塞大学-阿默斯特	

吴华强
清华大学

合作次数: 6

还有 71 个合作者